Disclosure Without Inference: LLM Agents Anchor on Disclosed Trade Data When Bargaining

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Abstract

As LLM agents begin to negotiate on behalf of people, the agents on either side of a trade may differ far more in capability than human negotiators do. One protection for the weaker side is an information floor: disclosing data it could not infer on its own, as US regulators did when publishing corporate bond trades, after which investors' trading costs fell. We test whether such a floor helps an LLM agent. In a calibrated bargaining game, an LLM client buys a Treasury note from a dealer with a private cost, and in some conditions also sees a summary of other dealers' reported purchases (the band). A Bayesian client facing the same dealer raises its share of the surplus by 11 percentage points with the band. The LLM client (Ornith-1.5-35B, reasoning off) showed no reliable gain and responded to the band by bidding its numbers rather than reasoning with them: a quarter of its counteroffers sat on the band's lowest price, where most were rejected. Its beliefs moved half as far as a Bayesian's and only weakly guided its bids. For this model, disclosure supplied information but not the inference needed to use it. This suggests that protocols for agent negotiation may need to test agents' capabilities, not only what they are shown.

1 Introduction

People are beginning to let LLM agents negotiate on their behalf. Unlike two human negotiators, the agents on either side of a trade can differ widely in capability: a dealer's agent may be far better than a client's at inferring what the other side will accept [1]. Markets have long protected the weaker side with disclosure. When US regulators began publishing corporate bond trades through TRACE, investors' trading costs fell [3]. We test it in a negotiation where a client agent buys an off-the-run US Treasury note from a single dealer with a private cost. In some conditions the client also sees a summary of other dealers' reported purchases, which we call the band. A Bayesian client facing the same dealer raises its share of the surplus from 62% to 73% with the band. The LLM client (Ornith-1.5-35B, with reasoning off) fell 18 points short of that gain (95% CI [4.8, 32.0]). Its behaviour points to a failure of inference: it bid the numbers it was shown rather than updating on them.

We make three contributions:

(1) **A testbed with a known value of disclosure.** A bargaining game calibrated to Treasury price volatility, with a Bayesian client that fixes exactly how much the band is worth to a client that uses it well.

(2) **Disclosure did not help the LLM client.** The band did not reliably improve its share of the surplus, and the result held on 20 unseen scenarios.

(3) **Without reasoning, the LLM client anchors rather than infers.** By eliciting the LLM's beliefs, we show that its bids land on disclosed numbers, its beliefs move half as far as a Bayesian's, and it over-weights weak evidence while underweighting strong evidence. Presentation and elicitation both act as anchors, so disclosure alone may not protect a weaker agent. We recognize that deployed agents are likely to reason step by step, so we treat this as evidence of an inference gap in this mode and not a general limitation of LLMs.

2 Background

Bargaining with private values. When both sides of a trade hold private values, no bargaining procedure can guarantee that every worthwhile trade happens while giving both sides an incentive to be truthful [7]. How the surplus is split then depends on the protocol and on what each side knows. Information design studies how a third party can change outcomes by choosing what to reveal [5].

LLM agents in bargaining. Lewis et al. [6] trained negotiation agents end to end. Qian et al. [8] compared humans, LLMs and Bayesian agents in a multi-player trading game and found that similar surplus can hide different strategies: LLMs made concessionary offers that were usually accepted. Like them, we use induced values and a computed benchmark, but we hold the agent fixed and vary its information.

Bayesian theory of mind and anchoring. Inferring another agent's private state from its actions can be modelled as Bayesian inverse planning [2, 10]. A client able to read a dealer's cost from its quote would need no disclosure. People, by contrast, often anchor on salient numbers [9], and LLMs show similar biases [4]. Our design distinguishes Bayesian use of a disclosure from anchoring on it.

3 Bargaining Game with a Bayesian Benchmark

A client buys \$10M of an off-the-run 10-year note from one dealer in three moves (Figure 1). First, the dealer quotes a price p. The client either accepts p, or makes a counter-offer $q < p$. The dealer accepts q if and only if $q \geq c$, its private cost.

The client has a private limit r, the most it will pay, and earns r minus the price, or zero if there is no trade. Both sides see an indicative evaluated price m. Our primary outcome is the client's share of the surplus, zero if there is no trade, or its capture as defined as follows:

$$\text{Capture} = \frac{r - \text{price}}{r - c}. \tag{1}$$

Calibration. Each scenario draws the dealer's cost c below m and the client's limit r above it, each at a random distance (LogNormal, median $1g$). Both sides know these distributions. The unit g, about \$3,400 on \$10M, is one tenth of the note's daily price volatility, which puts it within the range of real dealer margins.

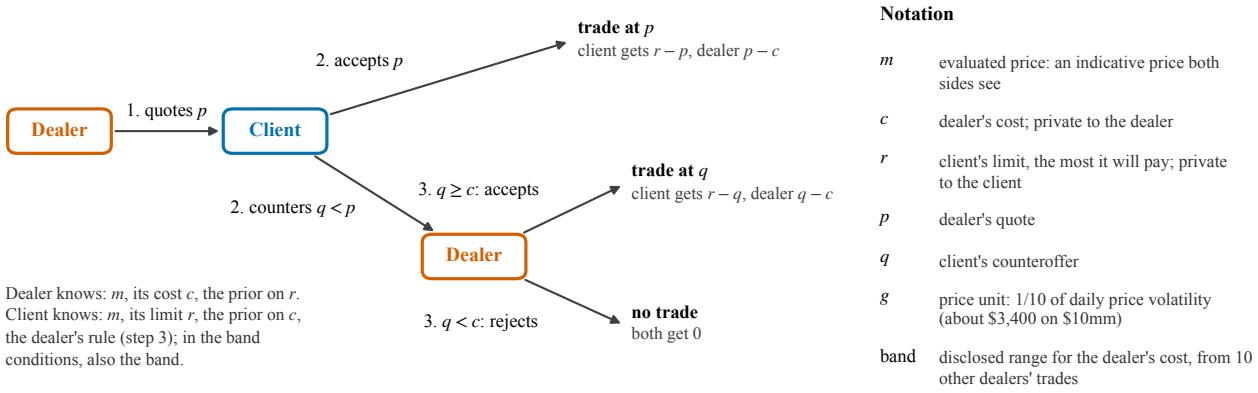


Figure 1: The bargaining game. The dealer quotes p; the client accepts or counters at $q < p$; the dealer accepts iff $q \geq c$, and its decision is final. A trade at x pays the client $r - x$ and the dealer $x - c$; no trade pays both zero. Each side knows its own private value and only the distribution of the other's.

Disclosure. The band summarises ten other dealers' reported purchases. Each is the market-wide typical cost plus noise, and this dealer's own cost differs from the typical cost by $d \sim \mathcal{N}(0, (0.25g)^2)$. The band's centre is therefore an unbiased estimate of this dealer's cost, typically off by $s = 0.356g$ (one standard deviation). Without the band, the client knows only the distribution of dealers' costs, whose standard deviation is $0.604g$, so the band is considerably more precise. When the band is disclosed, the client reads:

> "Reported purchases by other dealers of similar off-the-run notes yesterday (10 trades) averaged X below their evaluated prices. Applied to today's evaluated price, that implies a typical dealer cost of *avg*. Individual dealers' costs vary around the typical level, so this dealer's own cost is within [*low*, *high*] about two times in three."

The range is $avg \pm s$, which contains the true cost 68% of the time, so the statement is accurate.

Rational Client. As a benchmark, we compute the play of a Bayesian client. Suppose the client holds a belief $G(q)$, the probability that the dealer's cost is at or below q. Then its best counter q^* is the counter that maximises $(r - q)\,G(q)$, and the highest quote it should accept is $\theta = r - \max_q (r - q)\,G(q)$. It accepts the quote p if $p \leq \theta$, and otherwise counters at q^*: a lower counter means a better price but a higher chance of rejection. Without the band, G is the prior. With the band, G is the Bayesian posterior.

The Dealer. The same fixed algorithm in every condition. It chooses the quote that maximises its expected profit over the prior on the client's limit, assuming an uninformed client. It rounds the quote to 0.001 points, and accepts any counter at or above its cost. It quotes $m - 0.425g$ in 88% of scenarios, so the quote explains 18% of the variation in the cost, considerably little.

Value of disclosure. We play the rational client against this dealer at the true cost and limit. On our 50 test scenarios the band raises capture from 62.1% to 73.2%, a gain of 11.1 percentage points (11.4 points over 1,000 scenarios). This is the most a client that uses the band perfectly can gain.

4 Experiments

4.1 Design

Choosing the client. Before the experiment, we chose the client model with a short pilot on separate scenarios. The pilot checked basic competence, for example, that at least 90% of the model's counteroffers were below the quote and within its own limit. Our first choice, Qwen3.6-35B, failed this check (67%). Ornith-1.5-35B passed (92%) and was used throughout. It runs at temperature 0.5, so that repeated runs of a scenario can differ, and with reasoning off, so that we measure how it bargains without step-by-step working; we leave reasoning mode to future work.

The Experiment. We varied two things: whether the client sees the band, and whether we elicit its beliefs. Belief elicitation asks for the client's probability that the dealer's cost lies below each of nine prices, once before it sees the quote and once after, so that we can compare its beliefs with a Bayesian client's. From the nine answers we read off the client's best guess of the cost, the median of its stated belief. Each condition was played on the same 50 scenarios, twice each. The main test (the band raises the client's capture without belief elicitation) and its decision rule were pre-registered before any negotiation with the band. All confidence intervals are 95% bootstrap intervals over scenarios (10,000 resamples).

4.2 LLM Agent Bargaining Results without Disclosure

Without the band, the client almost made a counteroffer in 98–99% of negotiations, while a rational client usually accepted the quote. Its counteroffers were also low: on average $1.3g$ below the evaluated price. About half were rejected, so the client kept only 36–41% of the surplus.

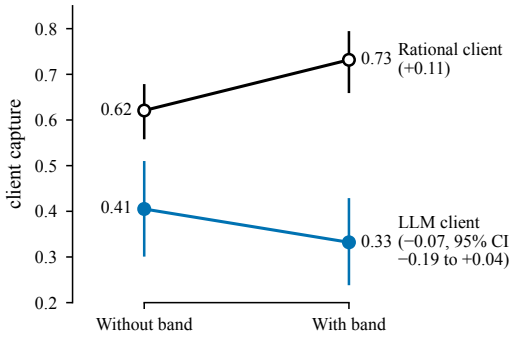


Figure 2: Client capture without and with the band. Rational client: open markers; LLM client, without belief elicitation: filled markers. The band raises the rational client's capture by 11% of the surplus; the LLM's change is not significant.

The client's counteroffers sit on numbers it has been given. Without belief elicitation, 77 of its 98 counteroffers land on one of two prices from the briefing: the typical dealer cost, 1g below the evaluated price (about $3,400 on the $10M trade; 46 counteroffers), or the low end of the briefing's 80% range of costs, 1.9g below (31 counteroffers; Figure 3a).

Belief elicitation itself also supplied prices to bid. It asks for the probability that the dealer's cost is below each of nine listed prices, and with elicitation, half of the client's counteroffers sat exactly on one of them (1.25g, 1.5g or 2g below the evaluated price). Without elicitation, the client almost never bid these prices.

Its stated beliefs do not drive its bids. Its beliefs imply a counteroffer about 0.47g below the evaluated price, but they do not predict its actual counteroffers ($\rho = 0.08$). Its counteroffers also do not rise with its own limit, although a rational client with a higher limit bids higher to make a rejection less likely.

4.3 Evaluating if an Information Floor Helps

The band did not improve the LLM client's outcome. Without belief elicitation, the client kept 41% of the surplus without the band and 33% with it (Figure 2). The difference, −7.3 points (95% CI $[-19.0, 3.8]$), is not significant, so our main hypothesis, that the LLM client uses the band to improve its outcome, is not supported. A rational client facing the same dealer raises its share from 62% to 73%. An exploratory paired test (per scenario, the rational client's gain from the band minus the LLM's) shows the LLM fell 18.4 points short (95% CI $[4.8, 32.0]$, $p \approx 0.008$). With belief elicitation, the band again made no difference (−0.2 points, 95% CI $[-10.2, 9.7]$), and the shortfall, 11.3 points ($[-2.7, 25.2]$, $p \approx 0.12$), is not significant.

4.4 How the LLM Agent Used the Band

The LLM agent reacted to the band, but by bidding its numbers rather than reasoning with them. As predicted, its counteroffers moved towards the band's centre, by 0.16g without belief elicitation (95% CI $[0.08, 0.24]$) and 0.26g with it ($[0.19, 0.33]$).

An exploratory look in Figure 3 shows this. Without belief elicitation, 26% of counteroffers sat exactly on the bottom of the band's

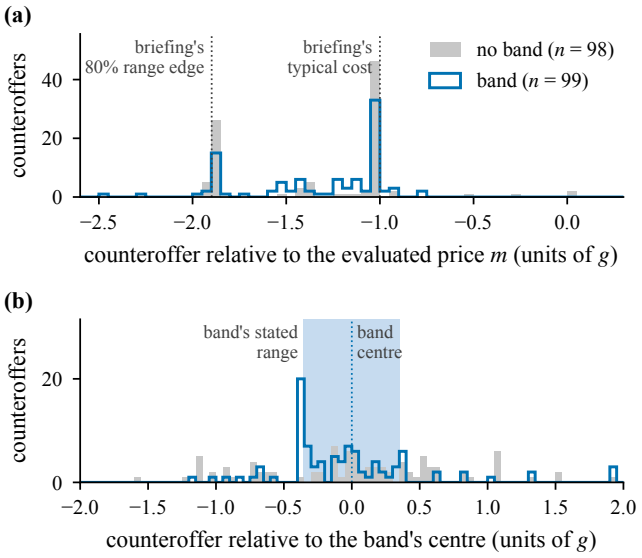


Figure 3: Where the client bids, without belief elicitation; grey bars: without the band, blue outline: with the band. (a) Counteroffers relative to the evaluated price: without the band they pile up on the briefing's numbers (dotted lines). (b) Counteroffers relative to the band's centre: with the band, many move onto the bottom of the band's stated range (shaded).

stated range, against 1% without the band. That low end is one standard deviation below the band's centre, so the dealer's cost lies above it about 84% of the time, and 77% of these counteroffers were rejected. A rejected bid earns the client nothing, so pulling a quarter of its bids onto this price is what makes the band's estimated effect on capture negative. The band did not replace the briefing's numbers: the typical cost from the briefing was still the client's most common counteroffer.

4.5 What the LLM Agent Believed

Belief elicitation shows that the band shifted the client's beliefs but did not make them more accurate. With the band, the client's best guess of the dealer's cost moved 0.23g towards the band's centre (95% CI $[0.14, 0.34]$), about half as far as a Bayesian client's (0.46g; Figure 4). Yet its belief as a whole was no closer to the Bayesian belief (change in mean squared error +0.012, 95% CI $[-0.004, 0.030]$). Its counteroffers, meanwhile, moved 0.25g closer to a rational client's (95% CI $[0.15, 0.33]$). Behaviour that moves towards the rational choice while beliefs stay no more accurate suggests anchoring rather than Bayesian use, at least without step-by-step reasoning.

The client also weighed evidence the wrong way round (Figure 4). A Bayesian moves its best guess in proportion to how much a piece of evidence reveals. The dealer's quote reveals little, because the dealer quotes the same price in most scenarios whatever its cost: even a client that knows the dealer's quoting rule moves its guess only 0.08g. Yet after seeing the quote, the client roughly halved its probability that the cost was at least 1g below the evaluated price

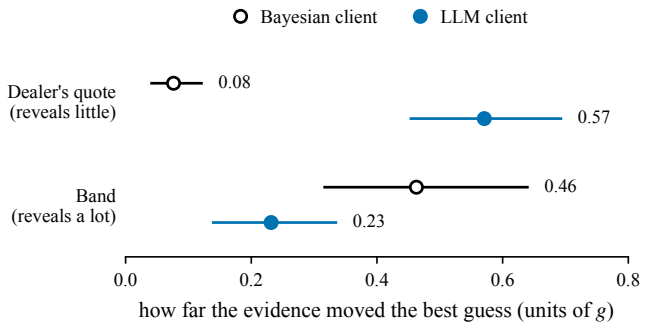


Figure 4: How far each piece of evidence moved the client's best guess of the dealer's cost, LLM against Bayesian, in the conditions with belief elicitation. Means over 50 scenarios with 95% CIs.

(from 0.71 to 0.36), moving its best guess $0.57g$. The band reveals far more, yet moved its guess only $0.23g$.

4.6 Confirmation from Unseen Scenarios

We repeated the experiment once on 20 scenarios never used during development, with the analysis code unchanged. The main findings held (Figure 5). The band again had no reliable effect on the client's share of the surplus (panel a): its confidence interval spans zero in both sets. They cannot test the shortfall, as the rational gain there is only 3.3 points. The client again moved its bids towards the band's centre (panel b), consistent with anchoring on the numbers it is given. And its beliefs again lagged the Bayesian client's (panel c): the band moved its best guess half as far as a Bayesian's in the main experiment, and not at all on the new scenarios.

5 Limitations and Future Work

Limitations. The game isolates the effect of information at the cost of realism. There is a single dealer, so the client has no competing quotes, its main protection in real markets; it makes one counteroffer, where real negotiations can go back and forth; and walking away is worth nothing, where a real client could trade elsewhere. The dealer is a fixed algorithm, not an agent: it neither adapts to nor exploits the client, and its quote reveals little about its cost, so we can detect a client over-reading the quote but not reading it skillfully. Costs, limits and reported trades are synthetic; only the price scale is calibrated to market data. Finally, we test one model, with reasoning off, under one prompt, so our results describe this model in this mode. This matters most for our interpretation: we attribute the failure to inference, and reasoning mode is the feature most likely to supply it. A deployed negotiating agent would more plausibly run with reasoning on, so our results may overstate the gap such an agent faces.

Future work. Future work should test whether the failure is specific to this setup. First, every client is told the distribution of dealers' costs, as parity with the Bayesian client requires, but the briefing states it as concrete prices, and these became the client's most common bids. Stating the same distribution without prices would show

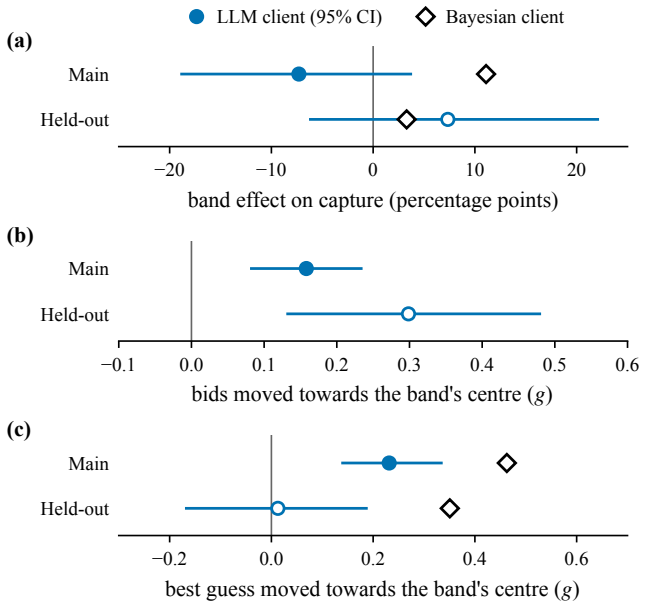


Figure 5: Main experiment (filled; 50 scenarios, two runs) against 20 held-out scenarios (open; one run), with 95% CIs. Diamonds: Bayesian client. (a) The band's effect on the client's share of the surplus. (b) How far bids moved towards the band's centre. (c) How far the client's best guess of the cost moved towards the band's centre.

whether the anchoring depends on being given a price. Second, and most directly, repeating the experiment with reasoning on would show whether the failure is one of inference that step-by-step working can fix, or whether the client anchors even when it reasons. Third, testing other model families and sizes would show whether any LLM agent can use an information floor to close a capability gap.

6 Conclusion

Disclosing other dealers' trade data did not reliably improve an LLM client's outcome, and without belief elicitation it delivered significantly less than a Bayesian client gains from the same information. The client did not ignore the disclosure; it treated the disclosed numbers as prices to bid. For this model with reasoning off, the floor supplied information but not the inference needed to use it. Whether reasoning, as a deployed agent would more likely use, closes this gap remains open.

Two lessons follow for protocol design. How information is presented matters as much as whether it is presented: an agent that anchors will bid whatever number it is shown, so a disclosure stated as a single price may do more harm than one stated as a range. Further, asking an agent about its beliefs is not neutral: the prices in our elicitation questions themselves became anchors. If agents cannot use what is disclosed to them, information floors alone cannot protect the weaker side, and protocols for agent negotiation may also need to bound the capability gap itself.

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